

Stochastic Concept Bottleneck Models

Reproduction questions

1. Can we reproduce the main empirical claim?

Dependency-aware SCBM should improve calibration and intervention efficiency without hurting prediction.

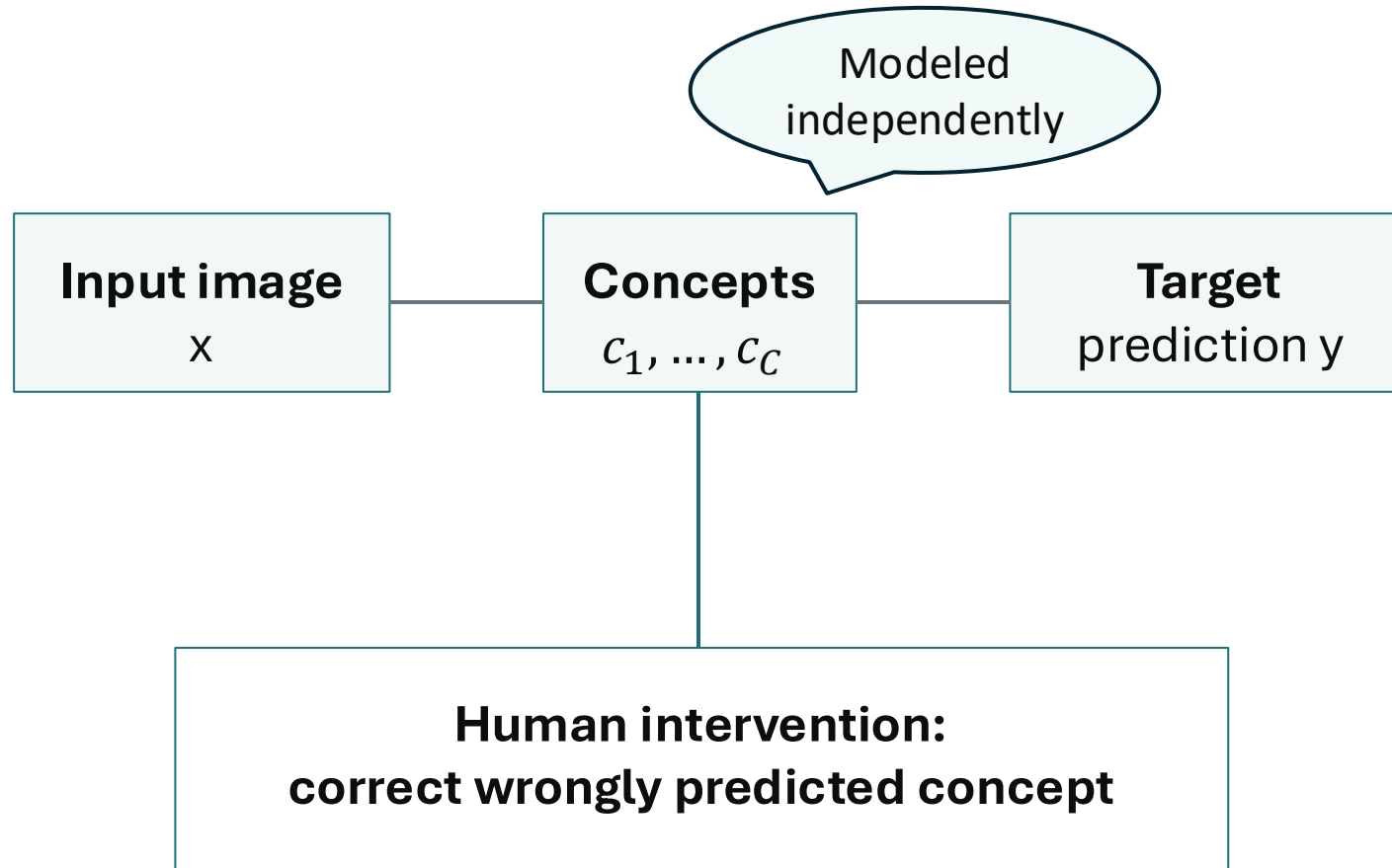
2. What is our contribution?

Refactored code, faster training, faster intervention evaluation.

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Limitation of Vanilla CBMs



Intervention Behavior

1. CBMS

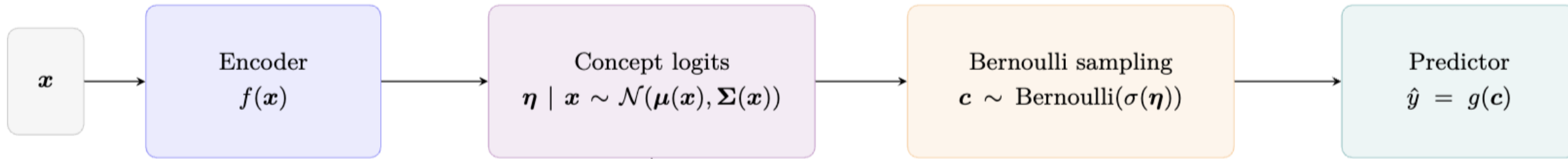
Intervening on one concept only changes that specific concept

2. SCBM

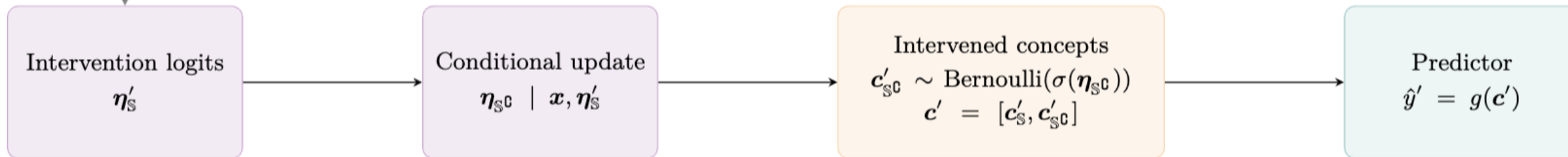
All concepts are updated via the learned concept logit distribution

SCBM replaces independent concept logits with a joint Gaussian bottleneck.

Prediction path

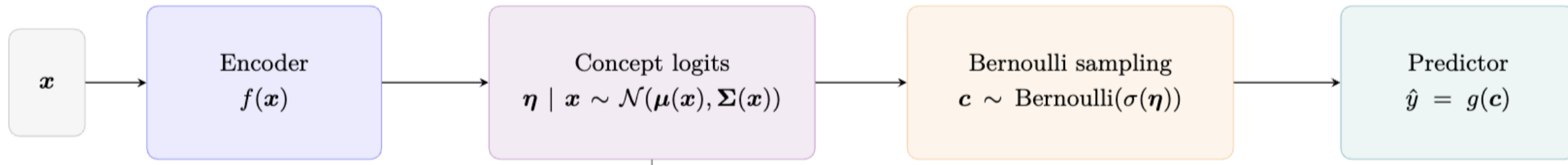


Intervention path



Training Objective

Prediction path



Concept Learning

$$\mathcal{L}_c = -\log p(\mathbf{c} \mid \mathbf{x}) = -\log \int p(\mathbf{c} \mid \boldsymbol{\eta}) p_\phi(\boldsymbol{\eta} \mid \mathbf{x}) d\boldsymbol{\eta}$$

$$\mathcal{L}_c \approx -\log \left(\frac{1}{M} \sum_{m=1}^M p(\mathbf{c} \mid \boldsymbol{\eta}^{(m)}) \right)$$

Target Learning

$$\mathcal{L}_y \approx -\log \left(\frac{1}{M} \sum_{m=1}^M p_\psi(y \mid \mathbf{c}^{(m)}) \right)$$

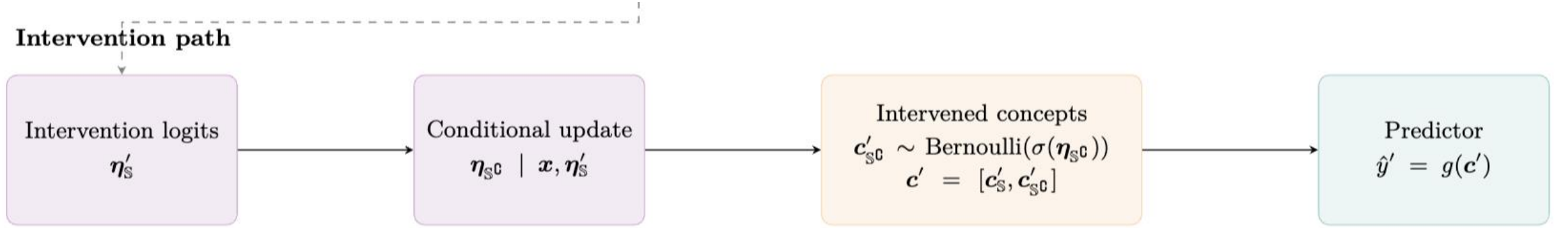
Precision Matrix Regularization

$$\mathcal{L}_{\text{reg}} = \sum_{i \neq j} \left| [\Sigma_\phi(\mathbf{x})^{-1}]_{ij} \right|$$

Final Objective

$$\mathcal{L} = \mathcal{L}_c + \lambda_y \mathcal{L}_y + \lambda_{\text{reg}} \mathcal{L}_{\text{reg}}$$

Dependency-Aware Intervention



Conditional Update

$$\eta_{\mathbb{S}^c} \mid \mathbf{x}, \eta'_{\mathbb{S}} \sim \mathcal{N}(\bar{\mu}_{\mathbb{S}^c}, \bar{\Sigma}_{\mathbb{S}^c \mathbb{S}^c})$$

$$\bar{\mu}_{\mathbb{S}^c} = \mu_{\mathbb{S}^c} + \Sigma_{\mathbb{S}^c, \mathbb{S}} \Sigma_{\mathbb{S}, \mathbb{S}}^{-1} (\eta'_{\mathbb{S}} - \mu_{\mathbb{S}})$$

$$\bar{\Sigma}_{\mathbb{S}^c \mathbb{S}^c} = \Sigma_{\mathbb{S}^c, \mathbb{S}^c} - \Sigma_{\mathbb{S}^c, \mathbb{S}} \Sigma_{\mathbb{S}, \mathbb{S}}^{-1} \Sigma_{\mathbb{S}, \mathbb{S}^c}$$

Intervention logits

$$\eta'_{\mathbb{S}} = \arg \max_{\eta_{\mathbb{S}}} \log p(\mathbf{c}'_{\mathbb{S}} \mid \eta_{\mathbb{S}})$$

$$\text{s.t.} \quad \begin{cases} \eta_i \geq \mu_i, & \text{if } c'_i = 1, \\ \eta_i \leq \mu_i, & \text{if } c'_i = 0, \end{cases} \quad \forall i \in \mathbb{S}, \quad \text{Correct direction}$$

$$(\eta_{\mathbb{S}} - \mu_{\mathbb{S}})^{\top} \Sigma_{\mathbb{S}, \mathbb{S}}^{-1} (\eta_{\mathbb{S}} - \mu_{\mathbb{S}}) \leq \chi^2_{|\mathbb{S}|, 1-\alpha} \quad \text{Remain plausible}$$

The reproduction tests SCBM across synthetic and natural-image datasets.

Dataset	Scale	Concepts	Why it matters
Synthetic	50,000	100 concepts	fixed latent covariance
CUB-200-2011	11,788	112 bird attributes	manual concepts
CIFAR-10	60,000	143 labels from GPT3	generated concepts

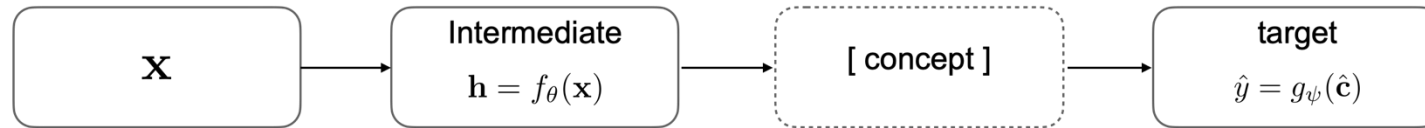
Evaluation dimensions

Prediction: concept accuracy and target accuracy

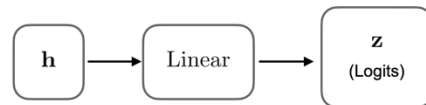
Calibration: Brier score and ECE

Intervention: accuracy after increasing concept corrections

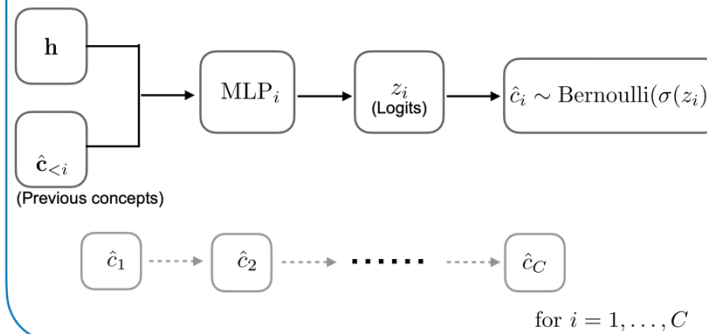
Compare SCBMs with 3 CBMs variants



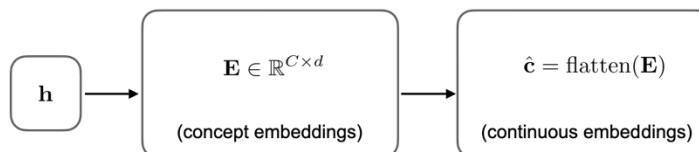
1. Soft CBM (Soft Bottleneck)



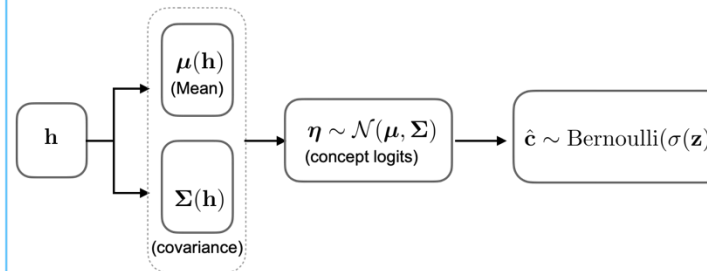
2. Hard CBM (Hard Bottleneck, Autoregressive)



3. CEM



4. SCBM



Methods

Hard CBM

Autoregressive CBM

CEM

Global SCBM

Amortized SCBM

SCBMs are competitive before intervention

Ours

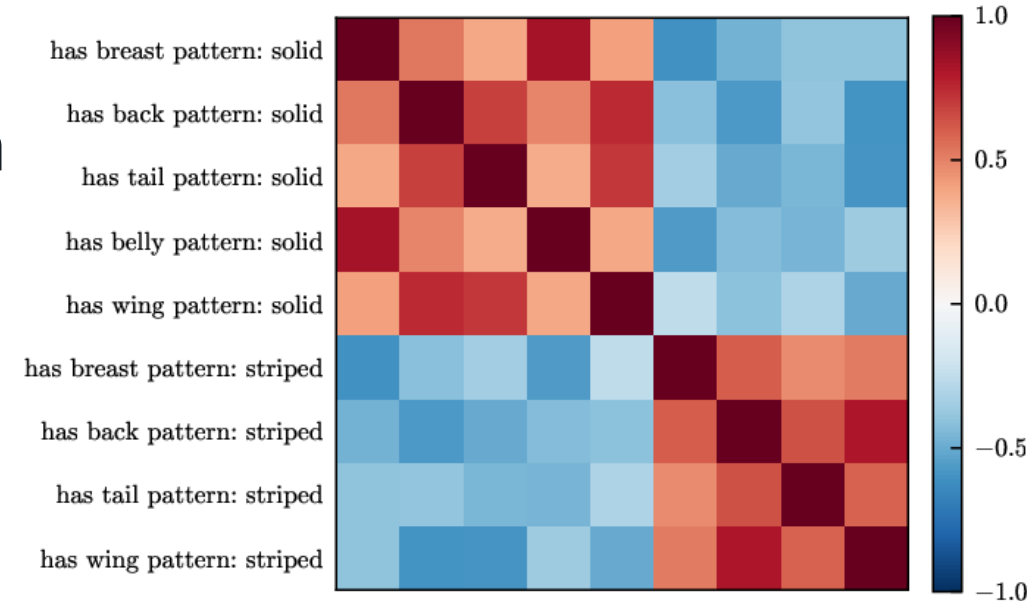
Dataset	Model	Concept Accuracy	Target Accuracy
Synthetic	Hard CBM	69.33 ± 0.07	66.76 ± 0.24
	CEM	69.37 ± 0.08	66.79 ± 0.37
	Autoregressive CBM	70.66 ± 0.03	67.28 ± 0.15
	Global SCBM	70.50 ± 0.04	<u>67.43 ± 0.20</u>
	Amortized SCBM	<u>70.53 ± 0.05</u>	67.47 ± 0.13
CUB	Hard CBM	94.83 ± 0.07	67.11 ± 0.68
	CEM	94.94 ± 0.09	68.79 ± 0.66
	Autoregressive CBM	95.24 ± 0.09	<u>68.83 ± 0.58</u>
	Global SCBM	94.79 ± 0.11	67.10 ± 0.76
	Amortized SCBM	<u>95.13 ± 0.10</u>	69.44 ± 0.59
CIFAR-10	Hard CBM	85.48 ± 0.07	69.72 ± 0.20
	CEM	85.23 ± 0.17	72.17 ± 0.61
	Autoregressive CBM	85.57 ± 0.10	69.56 ± 0.53
	Global SCBM	<u>85.87 ± 0.03</u>	70.89 ± 0.37
	Amortized SCBM	86.08 ± 0.07	<u>71.97 ± 0.26</u>

Original

Dataset	Method	Concept Accuracy	Target Accuracy
Synthetic	Hard CBM	61.42 ± 0.07	58.38 ± 0.39
	CEM	61.42 ± 0.12	58.01 ± 0.49
	Autoregressive CBM	<u>62.17 ± 0.11</u>	59.60 ± 0.62
	Global SCBM	61.57 ± 0.05	58.39 ± 0.53
	Amortized SCBM	62.41 ± 0.20	<u>58.96 ± 0.38</u>
CUB	Hard CBM	94.97 ± 0.07	67.72 ± 0.57
	CEM	95.12 ± 0.07	<u>69.60 ± 0.30</u>
	Autoregressive CBM	95.33 ± 0.07	69.24 ± 0.44
	Global SCBM	94.99 ± 0.09	68.19 ± 0.63
	Amortized SCBM	<u>95.22 ± 0.09</u>	69.87 ± 0.56
CIFAR-10	Hard CBM	85.51 ± 0.04	69.73 ± 0.29
	CEM	85.12 ± 0.14	72.24 ± 0.33
	Autoregressive CBM	85.31 ± 0.06	68.88 ± 0.47
	Global SCBM	<u>85.86 ± 0.04</u>	70.74 ± 0.29
	Amortized SCBM	86.00 ± 0.03	<u>71.66 ± 0.25</u>

SCBMs are competitive before intervention

Dataset	Model	Concept Accuracy	Target Accuracy
Synthetic	Hard CBM	69.33 ± 0.07	66.76 ± 0.24
	CEM	69.37 ± 0.08	66.79 ± 0.37
	Autoregressive CBM	70.66 ± 0.03	67.28 ± 0.15
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CIFAR-10	Hard CBM	85.48 ± 0.07	69.72 ± 0.20
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	Autoregressive CBM	85.57 ± 0.10	69.56 ± 0.53
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Interpretation

1. Global vs Amortized

flexibility of instance-wise covariance prediction

2. CEM on CIFAR-10

expressiveness-interpretability trade-off

SCBMs are better calibrated

For Brier score and ECE, lower is better.

Ours

Dataset	Model	Brier	ECE
Synthetic	Hard CBM	20.02 ± 0.04	5.89 ± 0.14
	CEM	20.08 ± 0.04	6.16 ± 0.09
	Autoregressive CBM	18.97 ± 0.02	2.34 ± 0.09
	Global SCBM	19.04 ± 0.02	<u>2.31 ± 0.07</u>
	Amortized SCBM	<u>19.01 ± 0.02</u>	2.11 ± 0.07
CUB	Hard CBM	4.02 ± 0.08	2.48 ± 0.13
	CEM	4.17 ± 0.08	3.34 ± 0.10
	Autoregressive CBM	<u>3.84 ± 0.07</u>	2.82 ± 0.10
	Global SCBM	4.01 ± 0.09	<u>2.41 ± 0.09</u>
	Amortized SCBM	3.70 ± 0.08	1.90 ± 0.09
CIFAR-10	Hard CBM	10.50 ± 0.06	5.29 ± 0.12
	CEM	10.84 ± 0.22	6.46 ± 0.55
	Autoregressive CBM	10.40 ± 0.12	5.23 ± 0.35
	Global SCBM	<u>9.98 ± 0.02</u>	<u>3.15 ± 0.07</u>
	Amortized SCBM	9.78 ± 0.05	1.96 ± 0.22

Original

Dataset	Method	Brier	ECE
Synthetic	Hard CBM	28.79 ± 0.09	22.38 ± 0.15
	CEM	29.32 ± 0.08	23.55 ± 0.09
	Autoregressive CBM	24.84 ± 0.32	13.54 ± 0.49
	Global SCBM	27.73 ± 0.09	20.10 ± 0.14
	Amortized SCBM	<u>25.58 ± 0.20</u>	<u>15.57 ± 0.55</u>
CUB	Hard CBM	3.93 ± 0.05	2.44 ± 0.06
	CEM	4.04 ± 0.05	3.25 ± 0.07
	Autoregressive CBM	<u>3.75 ± 0.05</u>	2.73 ± 0.05
	Global SCBM	3.87 ± 0.06	<u>2.33 ± 0.09</u>
	Amortized SCBM	3.64 ± 0.07	1.85 ± 0.08
CIFAR-10	Hard CBM	10.42 ± 0.05	4.93 ± 0.17
	CEM	11.06 ± 0.16	7.11 ± 0.39
	Autoregressive CBM	10.70 ± 0.05	6.07 ± 0.10
	Global SCBM	<u>9.95 ± 0.02</u>	<u>2.88 ± 0.11</u>
	Amortized SCBM	9.84 ± 0.02	2.22 ± 0.12

SCBMs are better calibrated

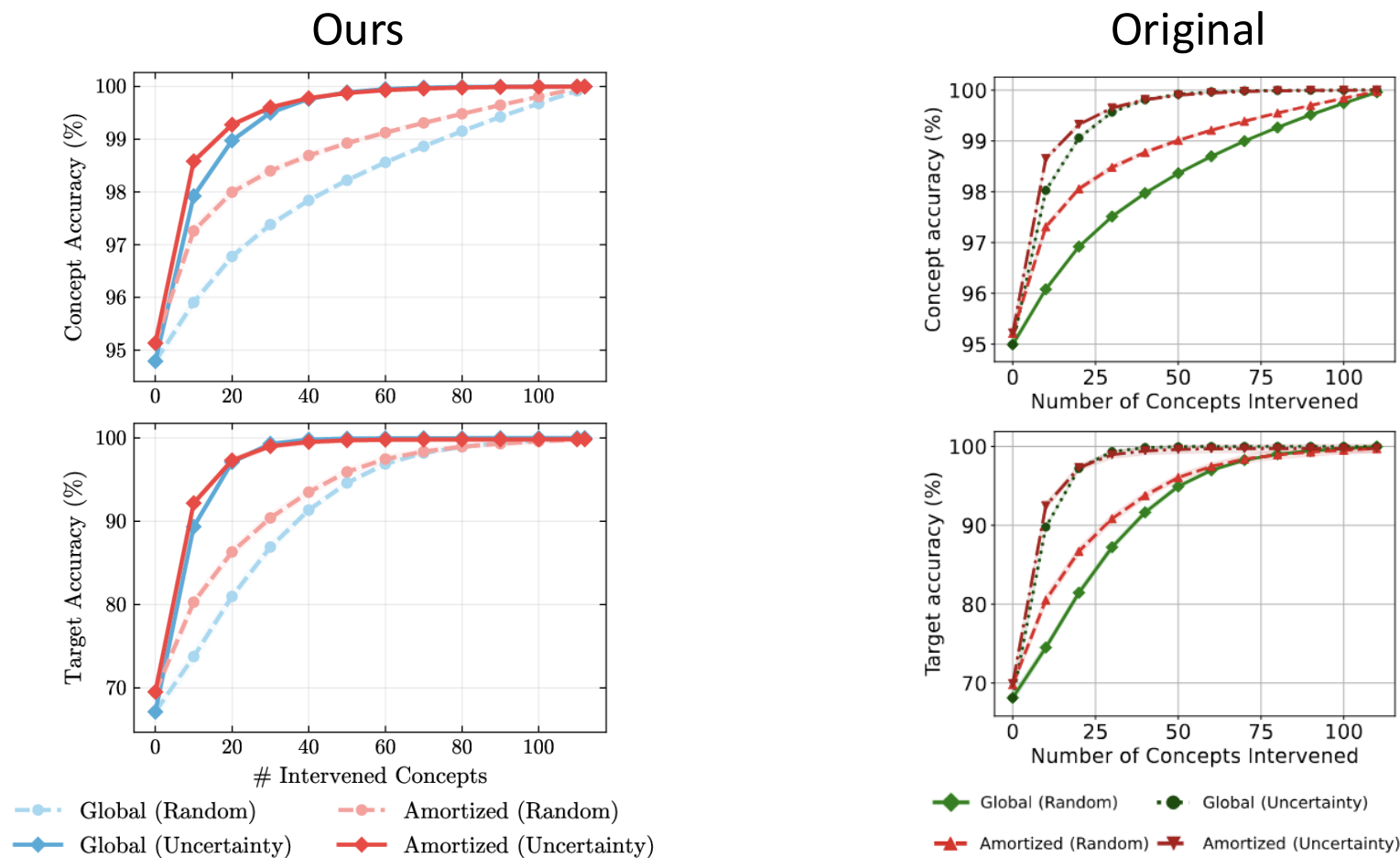
For Brier score and ECE, lower is better.

Dataset	Model	Brier	ECE
Synthetic	Hard CBM	20.02 ± 0.04	5.89 ± 0.14
	CEM	20.08 ± 0.04	6.16 ± 0.09
	Autoregressive CBM	18.97 ± 0.02	2.34 ± 0.09
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	CEM	10.84 ± 0.22	6.46 ± 0.55
	Autoregressive CBM	10.40 ± 0.12	5.23 ± 0.35
	Global SCBM	<u>9.98 ± 0.02</u>	<u>3.15 ± 0.07</u>
	Amortized SCBM	9.78 ± 0.05	1.96 ± 0.22

Why this matters for interventions

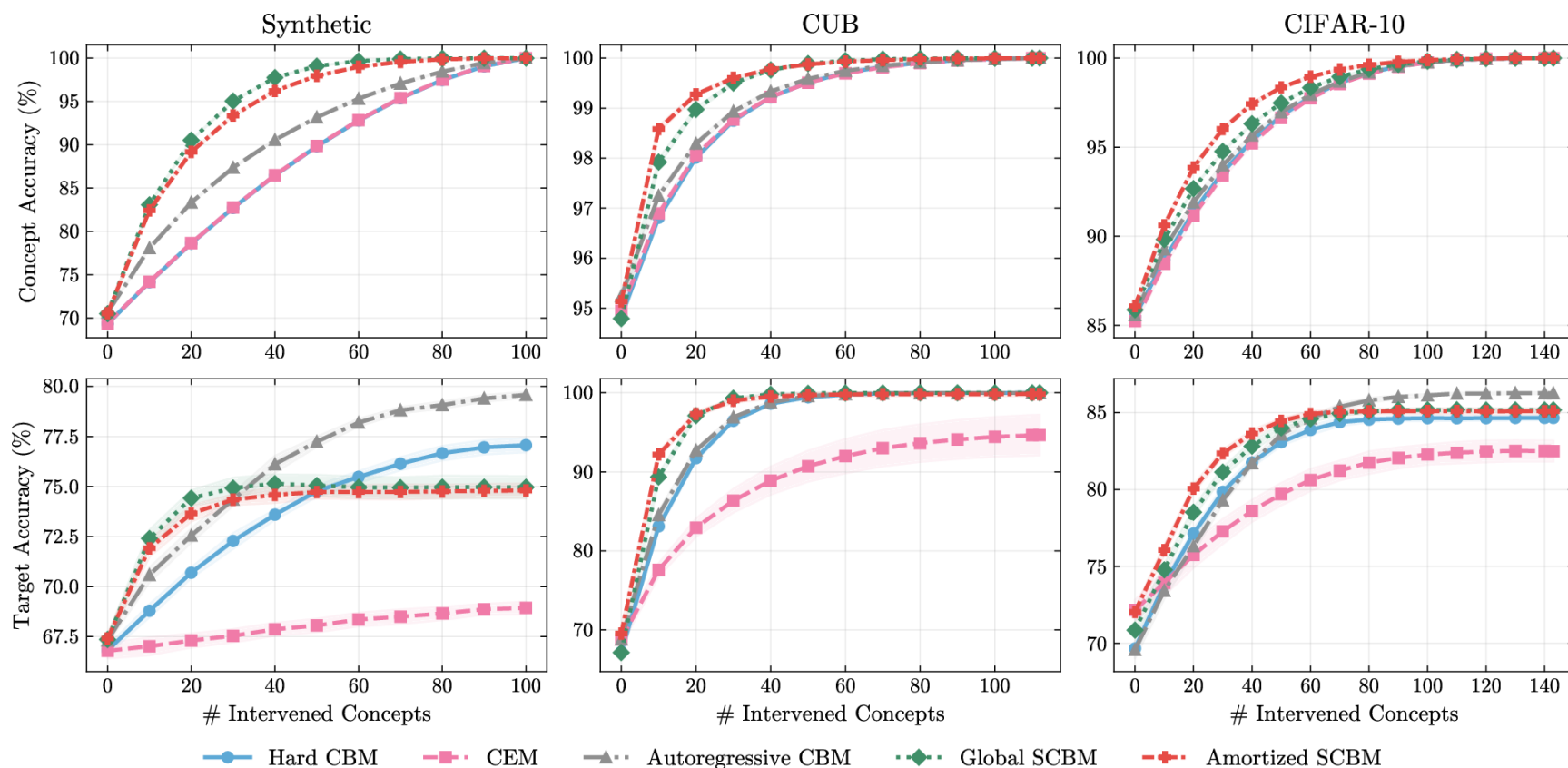
The uncertainty policy chooses concepts closest to probability 0.5. Better calibrated concept probabilities make those choices meaningful.

SCBM's predicted concept probabilities carry meaning uncertainty information



Policy comparison on CUB

SCBM variants arrive early, where concept interventions are practical.



Core reproduction

SCBM variants usually have steeper accuracy gains after few corrections.

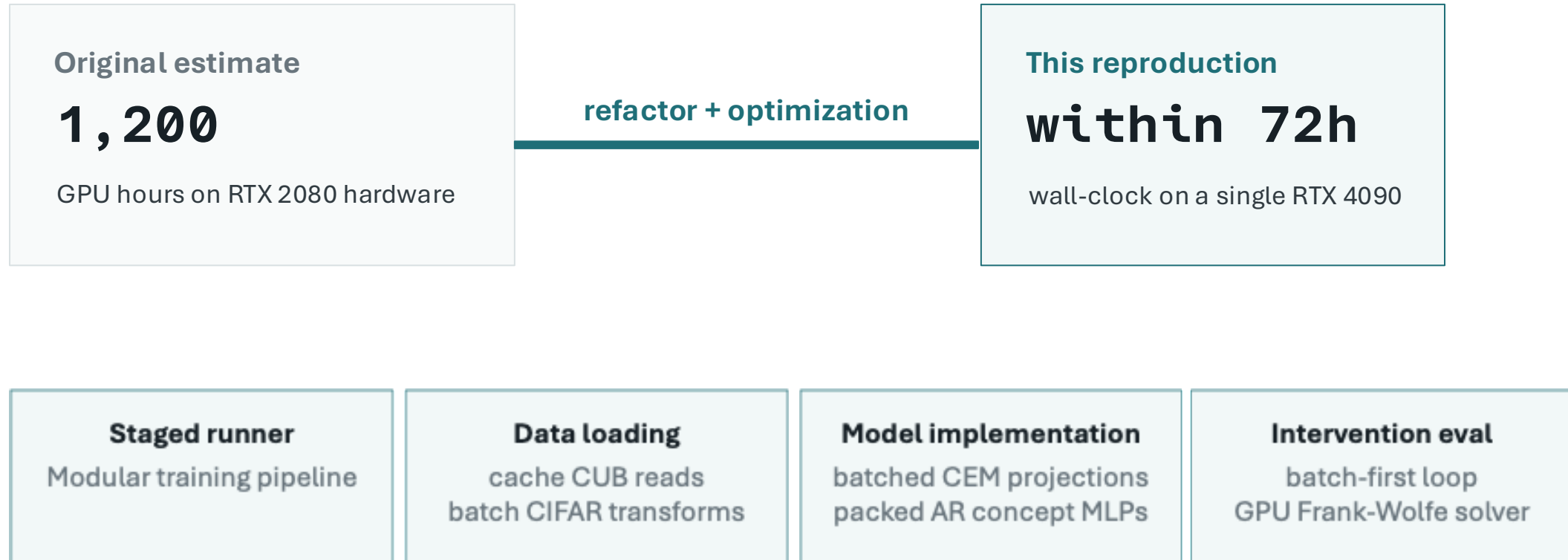
Mechanism

Correcting one concept updates correlated non-intervened concepts through the learned Gaussian.

Note

Autoregressive CBM can catch up at larger budgets; this is less aligned with the intended use case.

Our contribution: practical reproduction



Selected Speedup: exclude hardware difference

Table 3: Timing comparison between the legacy and optimized implementations.

Dataset	Model	Stage	Legacy (s)	Optimized (s)	Speedup
CIFAR-10	Autoregressive CBM	pretrain	188.2	8.7	21.6×
		concept	198.4	11.4	17.4×
		target	111.8	3.6	31.1×
		intervention	1264.6	86.1	14.7×
	CEM	joint	263.3	12.6	20.9×
		intervention	185.5	74.5	2.5×
	Hard CBM	joint	112.8	14.0	8.1×
		intervention	193.3	73.3	2.6×
	Amortized SCBM	joint	120.2	29.3	4.1×
		intervention	278.4	104.8	2.7×
	Global SCBM	joint	124.6	27.5	4.5×
		intervention	504.8	104.2	4.8×

The main SCBM result is reproducible

Method	Training	Inference
Hard CBM	5x	1x
CEM	5x	1x
Autoregressive CBM	5x	15x
Global SCBM	5x	1x
Amortized SCBM	5x	1x

Can we reproduce it?

Broadly yes.

SCBM is competitive before intervention, best calibrated across the three datasets, and stronger at low intervention budgets.

What did we contribute?

Feasible reproduction.

Refactored training, optimized model/data paths, batch-first interventions, and a validated GPU solver reduced practical cost.

What should not be overclaimed?

Runtime is sensitive.

SCBM's core value is dependency-aware intervention without a fixed autoregressive concept order.

Thanks

Q & A